MIT
MIVA OPEN UNIVERSITY

School of Post Graduate Studies

Master of Information Technology

40 Marks

2026/2027 Academic Session
First Semester Continuous Assessment

MIT 8103 Advanced Database Systems

Assessment Type: Individual Practical Database Portfolio

Submission Platform: LMS + GitHub

1. Overview

Students are required to develop and maintain an Advanced Database Systems Portfolio throughout the semester. The portfolio assesses their ability to design, implement, analyse, optimise, and manage advanced database systems in practical contexts.

This is not a conventional written assignment. Students must provide practical evidence of technical work, supported by appropriate explanations and documentation.

2. Database Case Study / Organisation Requirement

To remove ambiguity, students are NOT required to use one single compulsory company or organisation. Each student should select one realistic organisation or system as a case study and use it consistently across the portfolio activities.

The selected case study may be a real organisation, a simulated organisation, or a realistic business scenario. The organisation should have enough data and processes to support database design, SQL queries, transactions, NoSQL modelling, and distributed or cloud database discussion.

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Each student must state the selected organisation clearly at the beginning of the portfolio documentation. The statement should include the organisation name or scenario, the nature of the business/system, the users of the database, the major data to be stored, and the key operations the database will support.

Examples of acceptable case studies include, but are not limited to:

Case Study / Organisation Type	**Nature of Database**	**Possible Database Operations**
University / Student Result Portal	Students, courses, departments, lecturers, grades, registration records	Register courses, enter grades, compute GPA, generate transcript
Bank / Fintech Platform	Customers, accounts, transfers, deposits, withdrawals, transaction logs	Transfer funds, reverse failed transactions, audit account balances
E-commerce Platform	Products, customers, orders, payments, delivery records, product attributes	Add products, place orders, update stock, track delivery
Hospital / Clinic System	Patients, doctors, appointments, prescriptions, laboratory records	Book appointment, store diagnosis, retrieve patient history
Hotel Reservation System	Guests, rooms, bookings, payments, check-in/check-out records	Reserve rooms, update availability, process payment
Library Management System	Books, borrowers, loans, returns, fines, staff records	Borrow book, return book, calculate fine, search catalogue
Ride-hailing / Logistics System	Drivers, riders/customers, trips, locations, payments, delivery status	Match request, update trip status, track delivery
Social Media / Comment System	Users, posts, comments, likes, follows, notifications	Post comment, like content, retrieve user feed
Real-time Game Leaderboard	Players, scores, rankings, sessions, achievements	Update score, retrieve ranking, manage leaderboard

Students may also propose another case study, provided it is realistic and supports the required portfolio tasks.

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3. Portfolio Components

Component	Activity	Marks
Portfolio 1	Database Design and Modelling	7
Portfolio 2	Query Processing and Optimisation	7
Portfolio 3	Transactions and Concurrency	7
Portfolio 4	NoSQL and Advanced Data Models	7
Portfolio 5	Distributed and Cloud Database Exercise	6
Final Documentation	Portfolio documentation and technical reflection	6
Total		**40**

4. Expected Deliverables for Each Portfolio Activity

Portfolio	Expected Evidence
Portfolio 1: Database Design and Modelling	Chosen case study description, ER diagram, relational schema, keys, constraints, and normalisation explanation.
Portfolio 2: Query Processing and Optimisation	SQL queries, query results, execution plans, indexing/optimisation evidence, and brief performance analysis.
Portfolio 3: Transactions and Concurrency	Transaction examples showing commit, rollback, isolation, concurrency issue or control mechanism, and screenshots/output evidence.
Portfolio 4: NoSQL and Advanced Data Models	NoSQL model based on the same case study, document/key-value/column/graph example, implementation evidence, and explanation of suitability.
Portfolio 5: Distributed and Cloud Database Exercise	Distributed/cloud database discussion or setup based on the same case study, including replication, sharding, availability, consistency, or deployment evidence.
Final Documentation	A 2-3 page summary, GitHub link, selected organisation, screenshots/evidence, individual reflection, and references/AI-use declaration where applicable.

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5. GitHub Repository Requirements

Each student is required to create a private GitHub repository for the course. The repository must contain practical evidence of work throughout the semester, including where applicable:

I. SQL scripts
II. Database schemas
III. ER diagrams or database models
IV. NoSQL implementations
V. Query execution plans and performance results
VI. Screenshots, configuration files, and test results
VII. Technical documentation and README files

Important: Students are expected to commit to work progressively. Waiting until the deadline to upload all files may reduce the evidence of continuous portfolio development.

Suggested Repository Structure:

```
advanced-database-portfolio/
├── portfolio-1-database-design/
├── portfolio-2-query-optimisation/
├── portfolio-3-transactions-concurrency/
├── portfolio-4-nosql/
├── portfolio-5-distributed-cloud/
└── README.md
```

6. LMS Submission

Students are required to submit a 2-3 page documentation through the LMS. The documentation should describe all five portfolio activities, include the selected case study/organisation, and provide a link to the GitHub repository.

The GitHub repository must be accessible to the assessor before submission. The LMS submission should include supporting screenshots or evidence where required.

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7. Individual Work

This is an individual assessment. Students may discuss general concepts with colleagues, but the database implementation, scripts, analysis, documentation, and conclusions submitted in the portfolio must represent individual work.

Students may use development tools, documentation, textbooks, online resources, and AI tools as learning aids where permitted. However, each student remains responsible for understanding, verifying, and explaining everything submitted.

8. Technical Verification for Large Cohort

Because of the large class size, a compulsory viva or presentation for every student is not required. Verification will rely mainly on submitted technical evidence, structured documentation, and progressive GitHub activity.

I. Progressive GitHub commits showing development throughout the semester.
II. Clear README files explaining how each portfolio activity was implemented and tested.
III. Screenshots, outputs, execution plans, logs, or configuration files that support the submitted work.
IV. A short individual technical reflection describing design choices, challenges faced, and lessons learned.
V. The lecturer may select a small sample of students for brief technical verification where clarification is needed.
VI. Students with unclear, copied, inaccessible, or suspicious submissions may be invited for an individual verification discussion.

The purpose is to verify practical understanding and ownership without requiring all students to participate in viva or presentation.

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9. Marking Rubric

Criteria	Marks
Technical correctness	8
Database design quality	6
Practical implementation	8
Query and performance analysis	5
Application of advanced database concepts	5
Documentation and GitHub organisation	5
Technical reflection/evidence of individual understanding	3
Total	**40**

10. Assessment Expectations

I. Technical correctness
II. Quality of database design
III. Practical implementation
IV. Application of advanced database concepts
V. Query and performance analysis
VI. Problem-solving ability
VII. Quality of technical documentation
VIII. Evidence of independent work
IX. Ability to explain the submitted solution if selected for verification
X. Simply submitting screenshots or copied code without demonstrating understanding will not receive full marks.

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11. Academic Integrity and AI Use Declaration

All submissions must comply with the University's academic integrity requirements. Students must properly acknowledge sources, code, datasets, libraries, documentation, and other external resources used in their work.

Where AI tools are used, students must include a brief AI Use Declaration stating:

I. The AI tool used.
II. What the AI tool was used for.
III. How the output was reviewed, corrected, tested, or verified by the student.

Use of AI tools does not remove the student's responsibility for the accuracy, originality, and understanding of the submitted work.

12. Submission Checklist

I. The selected case study/organisation is clearly stated.
II. All five portfolio activities use the selected case study consistently, unless otherwise approved by the lecturer.
III. All five portfolio activities have been completed.
IV. The GitHub repository is properly organised.
V. Practical work and supporting evidence are included.
VI. README/documentation clearly explains the work.
VII. Required files have been submitted through the LMS.
VIII. The GitHub link works and can be accessed by the assessor.
IX. All sources, datasets, libraries, and external resources are properly referenced.
X. AI use, where applicable, has been declared.
XI. The student has included a clear individual technical reflection and can explain submitted work if selected for verification.

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13. Final Note

The goal is not merely to submit a collection of files but to build a portfolio that demonstrates the ability to apply advanced database concepts to real-world computing problems.

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